

White-Matter Hyperintensity (WMH) Segmentation

Progress Report Weeks 1-4

BME 4408 — Medical Imaging Course Project

Dataset: MICCAI 2017 WMH Segmentation Challenge found at
<https://doi.org/10.34894/AECRSD>

Overview

#	Requirement	Delivered
1	Skull stripping	✓ Week 2
2	Intensity normalisation	✓ Week 2
3	Bias field correction	✓ Week 2
4	WMH segmentation on FLAIR	✓ Week 3
5	Periventricular vs deep labelling	✓ Week 4
6	Lesion count	✓ Week 4
7	Total lesion volume (burden)	✓ Week 4
8	Largest lesion — diameter and volume	✓ Week 4
9	Spatial distribution across hemispheres	✓ Week 4
10	Mild / moderate / severe classification (bonus)	● Weeks 5–6
11	Benchmarked against the Challenge dataset	● Week 7

Week 1 - Data Acquisition

Downloaded and verified all 170 subjects

- Every file opened and checked: no corrupt data — **170/170 passed**

Organised the metadata

- Built a master index: voxel spacing, site, scanner, orientation and lesion statistics for every subject
- Froze the train/validation/test split before looking at any image

Built a synthetic test brain

- A fake volume with lesions at *exactly known* sizes and distances, so the code can be checked against an answer we chose
- This catches the four bugs real MRI cannot test for: wrong voxel maths, lesion counting rules, left/right flips, millimetres-vs-voxels

Two facts that shaped every later decision:

- **Slices are 3.0 mm apart while pixels within a slice are 0.56–1.30 mm apart.** This gives up to a 5.4× difference. Every distance and volume calculation has to account for this or it is silently wrong.
- **Two of the five test scanners never appear in training.** The dataset is designed to test whether a method learned *what a lesion looks like* or merely *what one hospital's scans look like*.

Week 2 - Preprocessing

Separated head from air — 3-class Otsu thresholding

- The standard 2-class Otsu failed on 1 patient (bright scalp pulled the cut-off above the brain, keeping only 25.6% of their lesions)
- **Final: 60/60 patients keep 100% of their lesions**

Corrected the scanner's uneven lighting — N4 at 2 levels

- Swept levels 2, 3 and 4 across 48 patients
- The software default destroys **up to 32% of lesion contrast** on the sickest patients
- At level 2 we remove about 14% of the uneven lighting
- We deliberately under-correct in exchange for never damaging lesions by >5%

Removed skull and scalp — HD-BET

- We tried HD-BET and deepbet, each on T1 and on FLAIR
- HD-BET beat deepbet although the difference was not statistically significant ($p = 0.339$)
- Final result: 100% of lesions retained on all 60 patients.

Week 2 - Preprocessing

K-means clustering ($k = 3$)

- On a T1 scan the three groups land naturally on cerebrospinal fluid, grey matter, and white matter, because those genuinely have different brightnesses.
- This is then used for intensity normalisation.

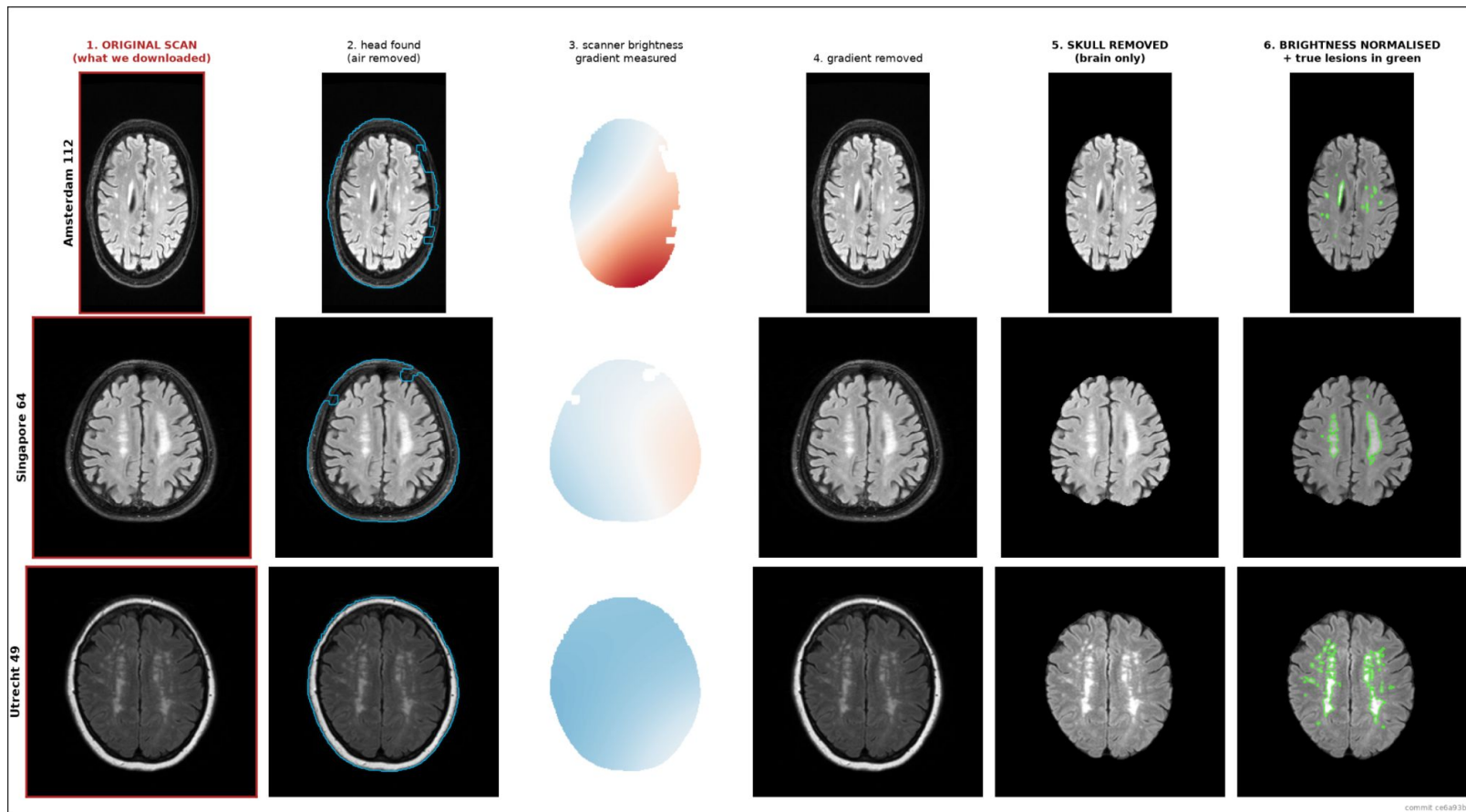
Intensity Normalisation — white-matter referencing

- Our method: find each patient's own normal-appearing white matter, take its median brightness, and divide the whole image by it.
- Healthy white matter = exactly 1.0 for every patient, so a lesion at 1.45 means "45% brighter than this patient's own normal tissue"

Tested three noise filters and rejected all three (Anisotropic diffusion, Gaussian blur and CLAHE)

- The standard quality measure (SSIM) rated 0.981, 0.993 and 0.79, while it had destroyed **399 lesions**
- So now instead of whole image metrics, we only measure metrics that directly talk about the lesions because that is what we care about

Week 2 - Preprocessing



Week 3 - Segmentation

First we built a simple baseline — brightness cut-off

- This gave us a Dice = **0.429**
- Now the neural network has something honest to beat

Then we trained a 2D U-NET

- 2 channels, one for FLAIR and the other for T1, because lesions are bright on FLAIR and dark on T1. That combination is far better than either alone
- This gave us a Dice = **0.806**

Then we ran 4 ensembles for the above and got a Dice = 0.808 (+0.003, which is within noise on 12 patients)

However, we were not sure if the model trained on recognizing anatomy or accidentally trained on recognizing a scanner

So we ran another test where we trained on 2 scanners and tested on the third (Leave one scanner out LOSO)

This **dropped the Dice to 0.738** meaning the model had learnt some scanner differences

Week 3 - Segmentation

The Solution: We used scanner simulating augmentation

We ran the LOSO test, training with the augmented data and this yielded a Dice=0.780
Clearly, augmentation helps, now our model is learning the anatomy of a lesion more than it did before.

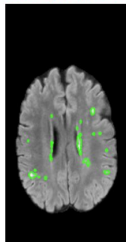
Then, we ran the 4 ensembles again with augmented data on the 48/12 split and got a Dice = 0.804
This is our models actual final performance.

It is what gives us an idea of how our model will perform on the 110 test patients in Week 7

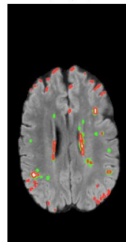
Metric	Value
Dice	0.8038
Lesion F1	0.7623
Hausdorff-95	4.39 mm
Absolute volume difference	22.10%

Week 3 - Segmentation

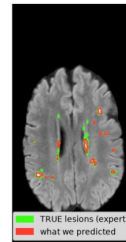
Amsterdam 100
TRUE lesions (what we must find)



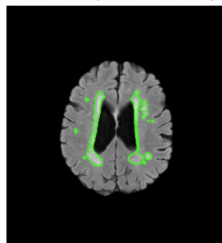
METHOD 1: brightness cut-off
(simple)



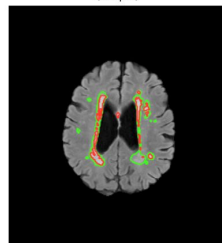
METHOD 2: four neural networks, combined
Dice 0.778



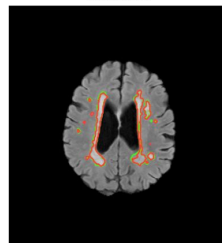
Singapore 56
TRUE lesions (what we must find)



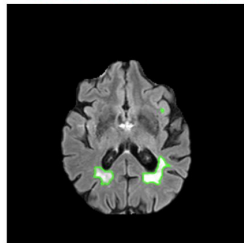
METHOD 1: brightness cut-off
(simple)



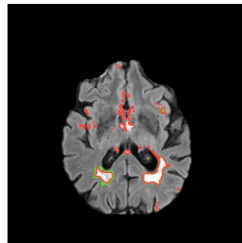
METHOD 2: four neural networks, combined
Dice 0.892



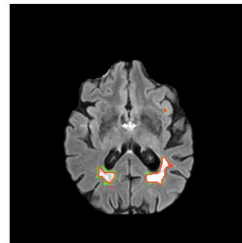
Utrecht 19
TRUE lesions (what we must find)



METHOD 1: brightness cut-off
(simple)



METHOD 2: four neural networks, combined
Dice 0.842



Week 4 - Labelling and Measurement

To label the lesions as periventricular / deep we first had to find and map out the ventricles. The distance from the ventricle edge will then be measured to perform the labelling.

We did the following:

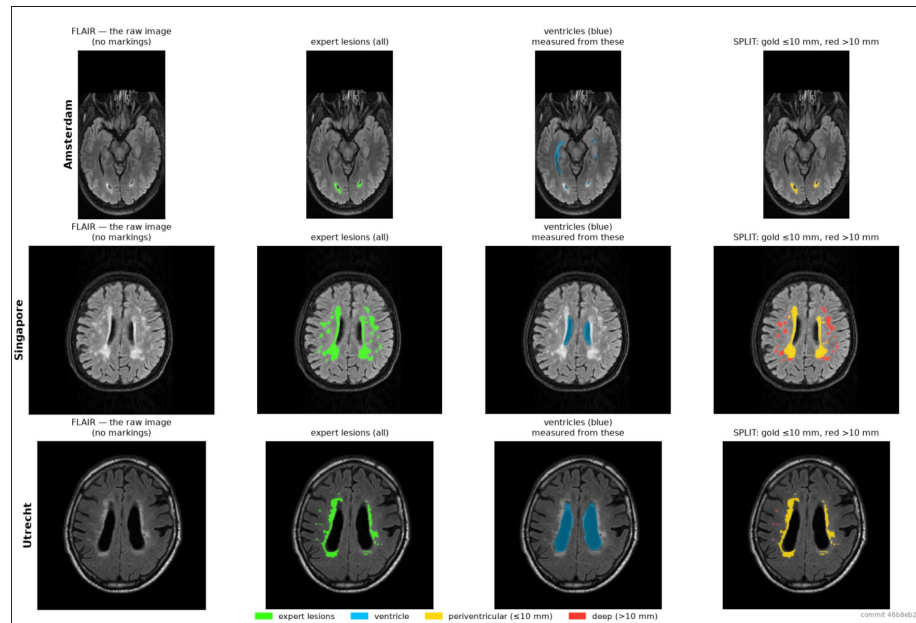
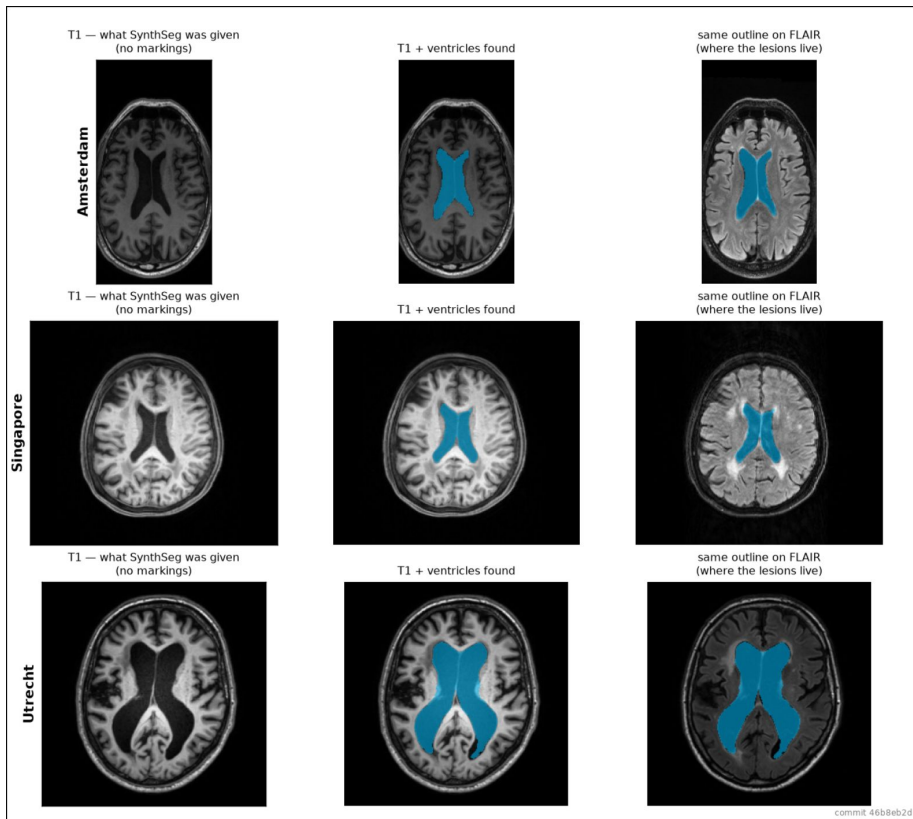
- FLAIR blacks out the fluid, so we tried to threshold the dark voxels, group them into connected blobs, keep the large ones near the brain's centre, close small gaps. All four worked on roughly **28% of patients**.
- **This did not work** because it leaned on the week 2 clustering, which was designed to accurately detect white matter but not CSF. Leaning on that was an error on our part.

The solution:

- SynthSeg (FreeSurfer): 60/60 patients, ~45 s each designed to accurately segment brain structures
- We measured on T1, Raw FLAIR and our processed FLAIR. T1 swallowed the least lesions, so we used it.

We then split every lesion into periventricular / deep based on a general 10mm rule. Voxels were not used due to the 3 mm in-slice difference.

Week 4 - Labelling and Measurement

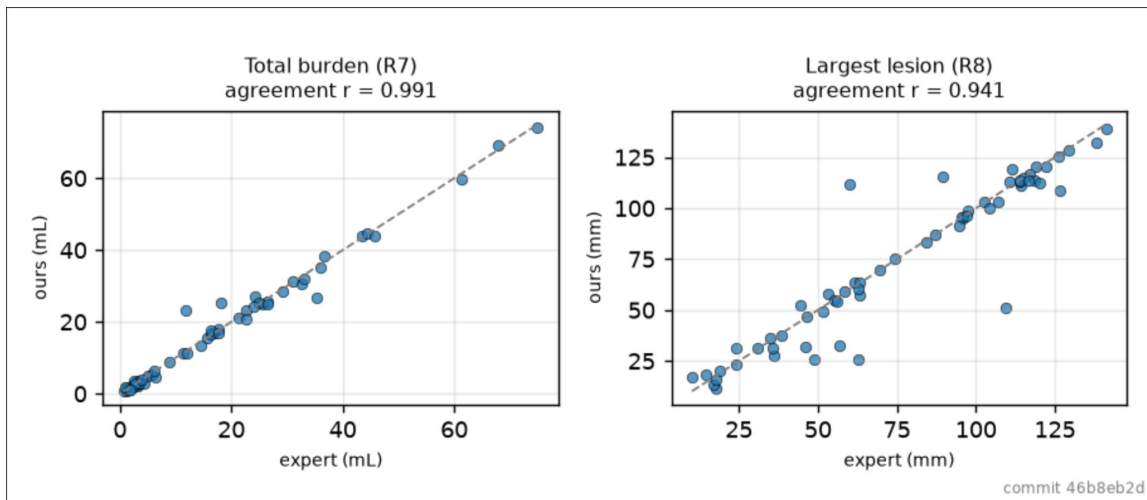


74.5% of the lesion burden is periventricular

Week 4 - Labelling and Measurement

We then measured the lesions for count, total burden, largest-lesion diameter and volume, left/right spread.

Every measurement **computed twice**, once from the expert masks, once from our model, so next week cannot train on its own guesses.



Agreement with the experts:

Metric	%
Burden	0.984
Count	0.946
Diameter	0.932
Largest Volume	0.956

We used Feret Diameter

Burden agrees more closely than count. We measure how much disease there is very accurately, and how many separate lesions slightly less accurately, because half of them are 5 voxels or smaller where a one-voxel disagreement merges or splits an object

Current Status and Next Steps

Weeks 1–4 are complete.

- The pipeline runs end to end: raw scan in, segmented and labelled lesions with quantitative features out. 92 automated phantom brain tests pass. The 110 official test subjects remain sealed.

Weeks 5–6 — Classification

- This is the ML / DL / statistical stage: categorise each subject as mild, moderate or severe WMH burden, using the lesion measurements produced in Week 4 (bonus).

Weeks 7–9

- Performance evaluation against the official leaderboard, consolidation of the work into one final document, and the final presentation.

The Team

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- Ahaan Shah
- Ishan Nathan

Thank You